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Data visualization with ggplot2 in R

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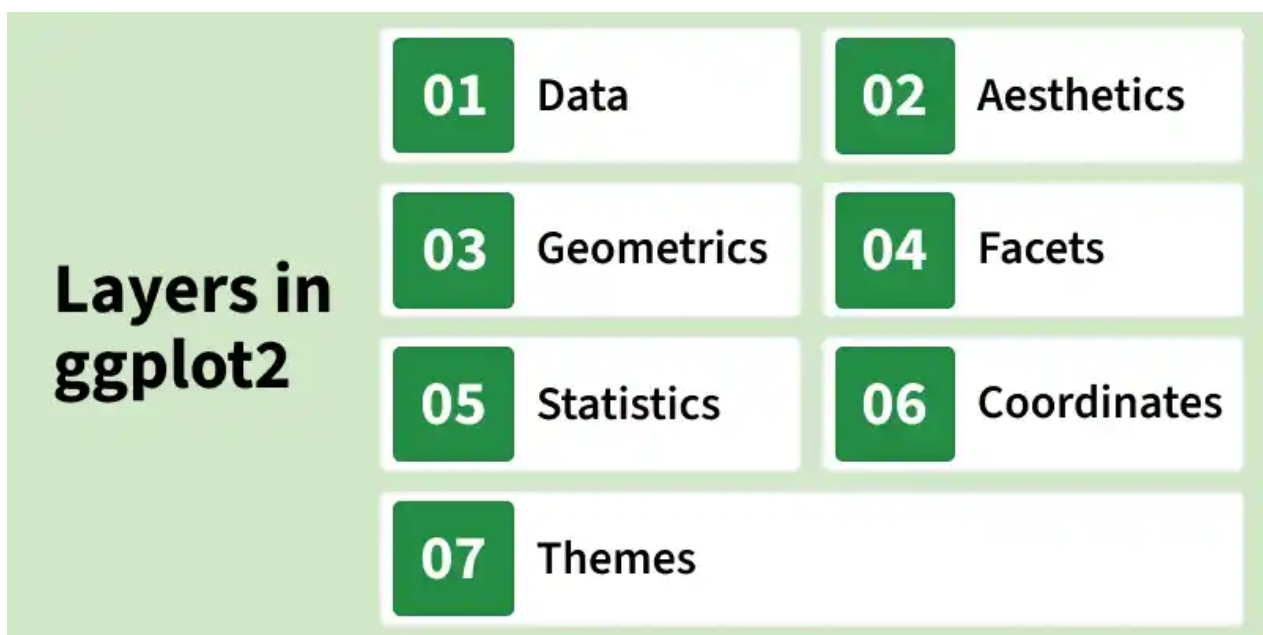


ggplot2 is a open-source data visualization package in R based on the concept of the Grammar of Graphics. It allows users to build complex and elegant visualizations by combining multiple layers in a structured way. Instead of writing long plotting code ggplot2 lets you construct graphs step by step using clear components.

- Based on the layered approach.
- Helps create clear, customizable and publication-quality visualizations.
- Widely used in data analysis, statistics and data science.

Layers of the Grammar of Graphics in ggplot2

ggplot2 builds every plot using multiple layers. Each layer has a specific role in defining how the data is displayed.



Layers

1. Data

The data layer represents the dataset used to create the visualization. It can be a data frame or any structured dataset in R.

- Specifies the source of the data.
- Can be defined globally or separately for each layer.

2. Aesthetics (aes)

The aesthetic layer maps variables in the dataset to visual properties of the plot.

- Common aesthetics: x, y, color, fill, size, shape, alpha, linetype.
- Controls how data values appear visually.

3. Geometric Objects(Geoms)

Geoms define how the data is displayed on the plot. Each geom represents a different visual representation of the data.

- **geom_point()**: Scatter plot
- **geom_line()**: Line plot
- **geom_bar()**: Bar chart
- **geom_histogram()**: Histogram
- **geom_boxplot()**: Boxplot

4. Facets

Faceting divides data into subsets and displays multiple plots based on categories. Useful for comparing groups within the same dataset.

- **facet_wrap()**: Wraps plots into multiple panels
- **facet_grid()**: Arranges plots in rows and columns

5. Statistics (Stats)

The statistical layer performs transformations on the data before plotting.

- Binning (used in histograms)
- Smoothing (e.g., regression lines using `geom_smooth()`)
- Summary statistics (mean, count, etc.)

Some geoms automatically apply statistical transformations.

6. Coordinates

The coordinate system controls how data points are positioned in space. Coordinates determine the relationship between data and the display area.

- **coord_cartesian()**: Cartesian coordinates
- **coord_fixed()**: Fixed aspect ratio
- **coord_polar()**: Polar coordinates

7. Themes

The theme layer controls the non-data elements of the plot (appearance). Themes improve readability and presentation quality.

- Background style

- Grid lines
- Fonts and text size
- Legend position

Examples: `theme_minimal()`, `theme_classic()`, `theme_bw()`

Step by Step Implementation

We will use the `mtcars`(motor trend car road test) dataset which is a built in dataset in R. It comprise fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

Step 1: Install and Load Required Packages

Install and load the necessary libraries. These packages provide tools for data manipulation and visualization. The `head()` function displays the first six rows of the dataset.

```
install.packages("vctrs")
install.packages("dplyr")

library(dplyr)
library(ggplot2)

head(mtcars)
```



Output:

A data.frame: 6 × 11

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

dataset

Now we print the summary of `mtcars` dataset using `summary` function.

```
summary(mtcars)
```



Output:

mpg	cyl	disp	hp
Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0
1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5
Median :19.20	Median :6.000	Median :196.3	Median :123.0
Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7
3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0
Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0
drat	wt	qsec	vs
Min. :2.760	Min. :1.513	Min. :14.50	Min. :0.0000
1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000
Median :3.695	Median :3.325	Median :17.71	Median :0.0000
Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375
3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000
.. : ..	.. : ..	.. : ..	.. : ..

Max.	:4.930	Max.	:5.424	Max.	:22.90	Max.	:1.0000
	am		gear		carb		
Min.	:0.0000	Min.	:3.000	Min.	:1.000		
1st Qu.:	0.0000	1st Qu.:	3.000	1st Qu.:	2.000		
Median	:0.0000	Median	:4.000	Median	:2.000		
Mean	:0.4062	Mean	:3.688	Mean	:2.812		
3rd Qu.:	1.0000	3rd Qu.:	4.000	3rd Qu.:	4.000		
Max.	:1.0000	Max.	:5.000	Max.	:8.000		

Summary of the data

Step 2: Data Layer

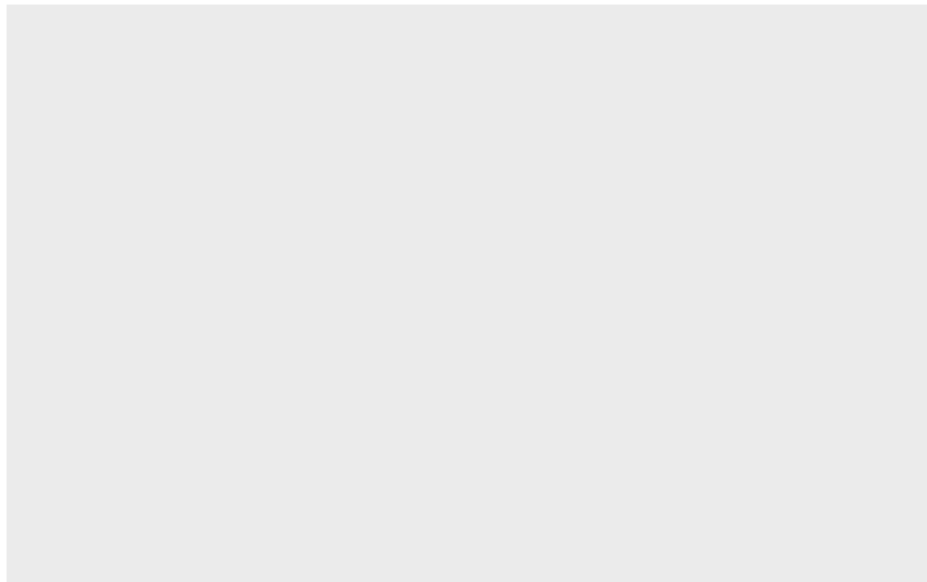
The data layer we define the source of the information to be visualize, let's use the mtcars dataset in the ggplot2 package.

```
ggplot(data = mtcars) +  
  labs(title = "MTCars Data Plot")
```



Output:

MTCars Data Plot



ggplot2 in R

Step 3: Aesthetic Layer

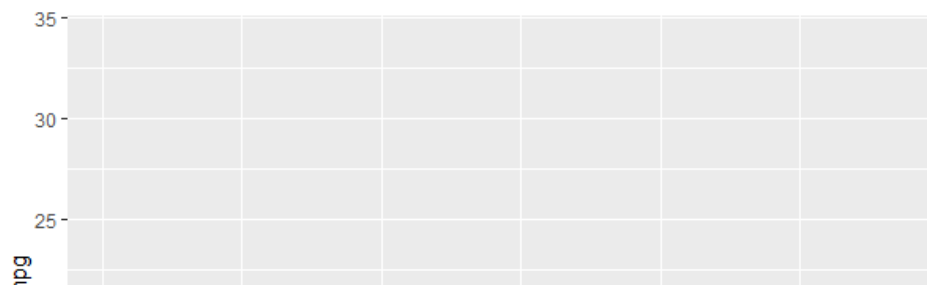
Here we will display and map dataset into certain aesthetics. Map horsepower (hp) to the x-axis, miles per gallon (mpg) to the y-axis and displacement (disp) to color.

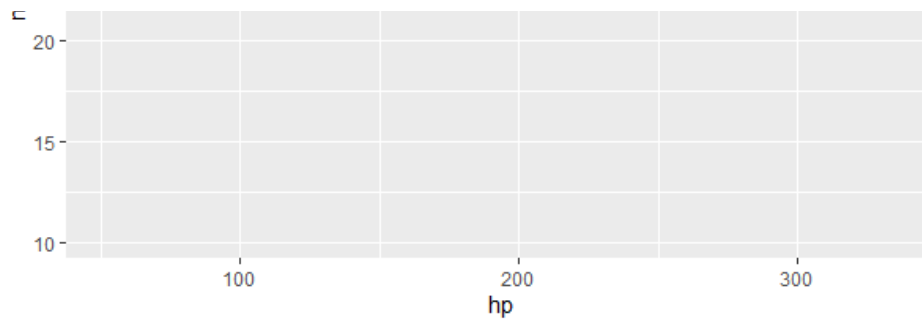
```
ggplot(data = mtcars, aes(x = hp, y = mpg, col = disp))+  
  labs(title = "MTCars Data Plot")
```



Output:

MTCars Data Plot





ggplot2 in R

The plot area is prepared with mapped aesthetics, but no points appear because no geometry layer is added.

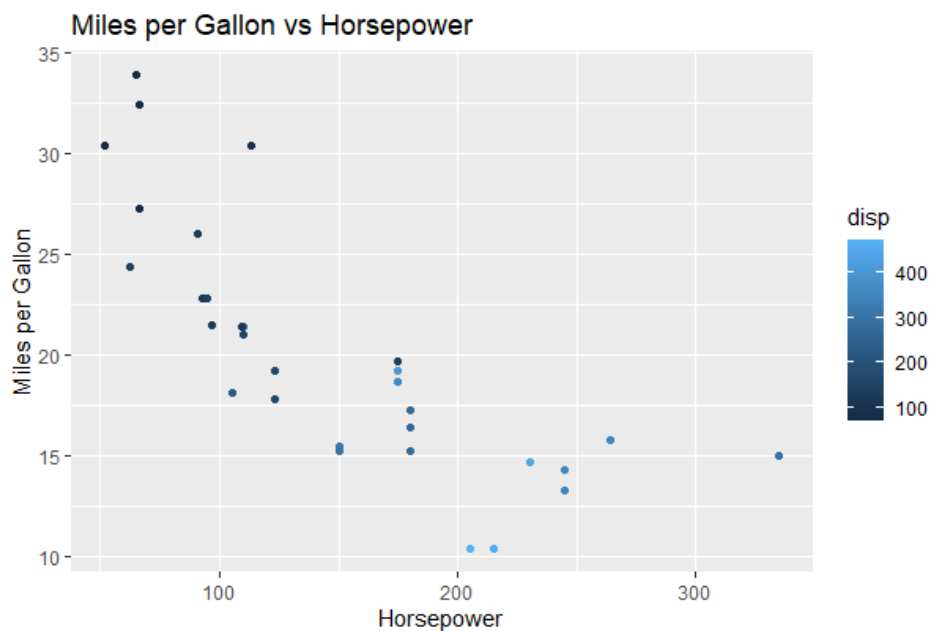
Step 4: Geometric layer

The geometric layer control the essential elements, see how our data being displayed using point, line, histogram, bar, boxplot.

```
ggplot(data = mtcars, aes(x = hp, y = mpg, col = disp)) +  
  geom_point() +  
  labs(title = "Miles per Gallon vs Horsepower",  
        x = "Horsepower",  
        y = "Miles per Gallon")
```



Output:



Data visualization with R and ggplot2

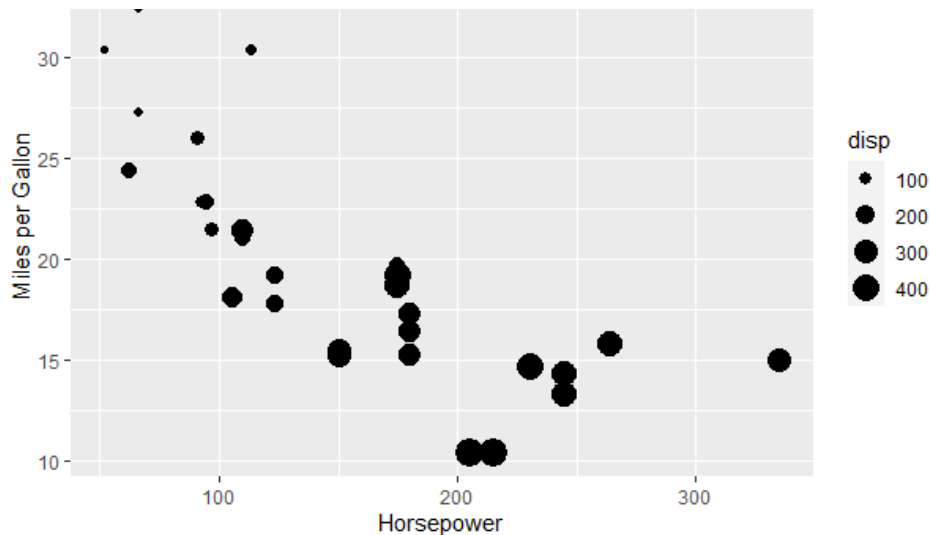
Add Size Aesthetic: Maps engine displacement (disp) to point size, so cars with larger engines appear as bigger dots in the scatter plot.

```
ggplot(data = mtcars, aes(x = hp, y = mpg, size = disp)) +  
  geom_point() +  
  labs(title = "Miles per Gallon vs Horsepower",  
        x = "Horsepower",  
        y = "Miles per Gallon")
```



Output:





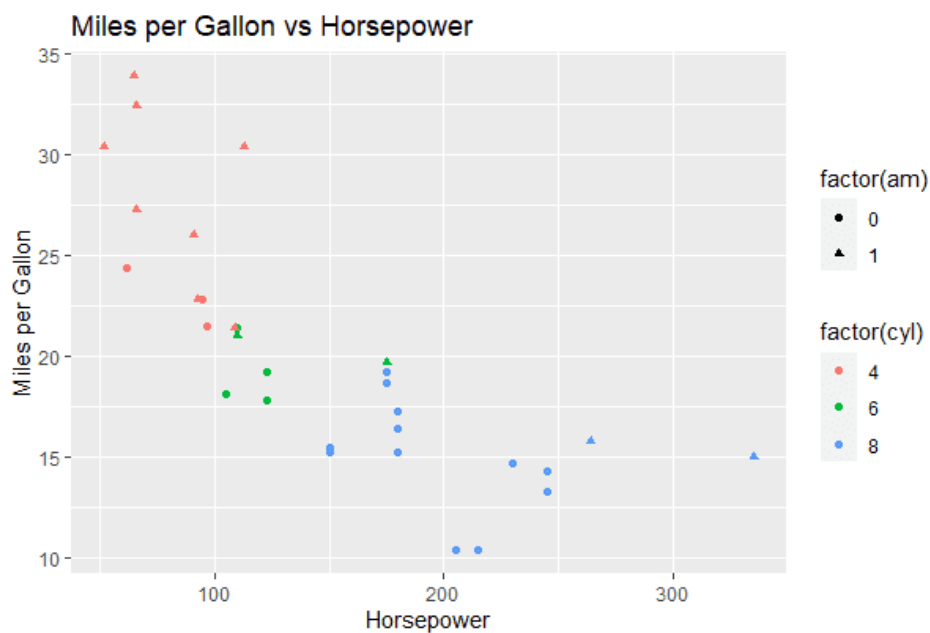
Data visualization with R and ggplot2

Add Shape and Color Categories: Uses color for cylinder count and shape for transmission type to visually differentiate car categories in one plot.

```
ggplot(data = mtcars, aes(x = hp, y = mpg, col = factor(cyl),
  shape = factor(am))) +
  geom_point() +
  labs(title = "Miles per Gallon vs Horsepower",
    x = "Horsepower",
    y = "Miles per Gallon")
```



Output:



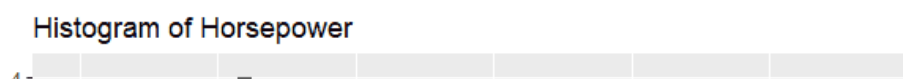
Data visualization with R and ggplot2

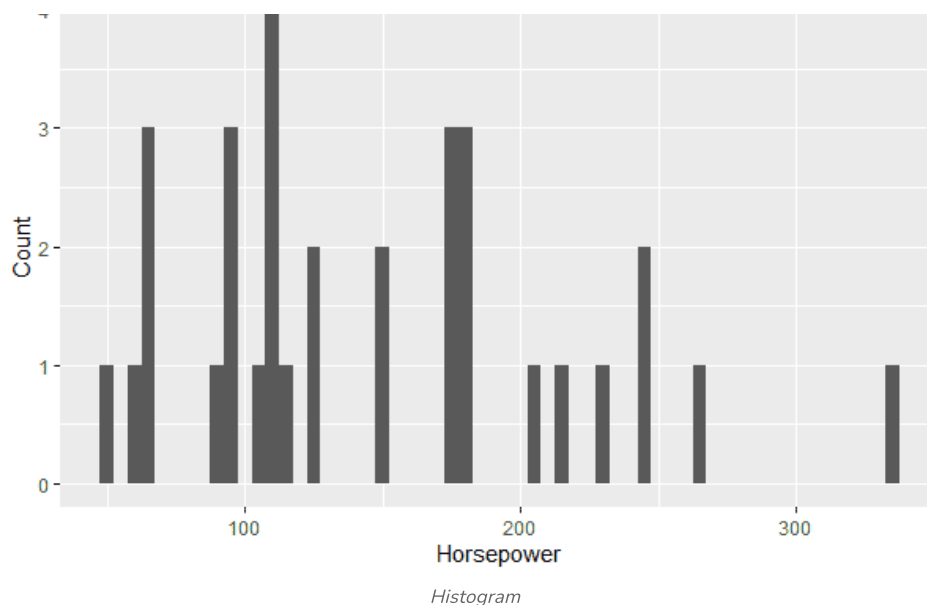
Histogram: Creates a histogram of horsepower to show its frequency distribution across different value ranges.

```
ggplot(data = mtcars, aes(x = hp)) +
  geom_histogram(binwidth = 5) +
  labs(title = "Histogram of Horsepower",
    x = "Horsepower",
    y = "Count")
```



Output:





Step 5: Facet Layer

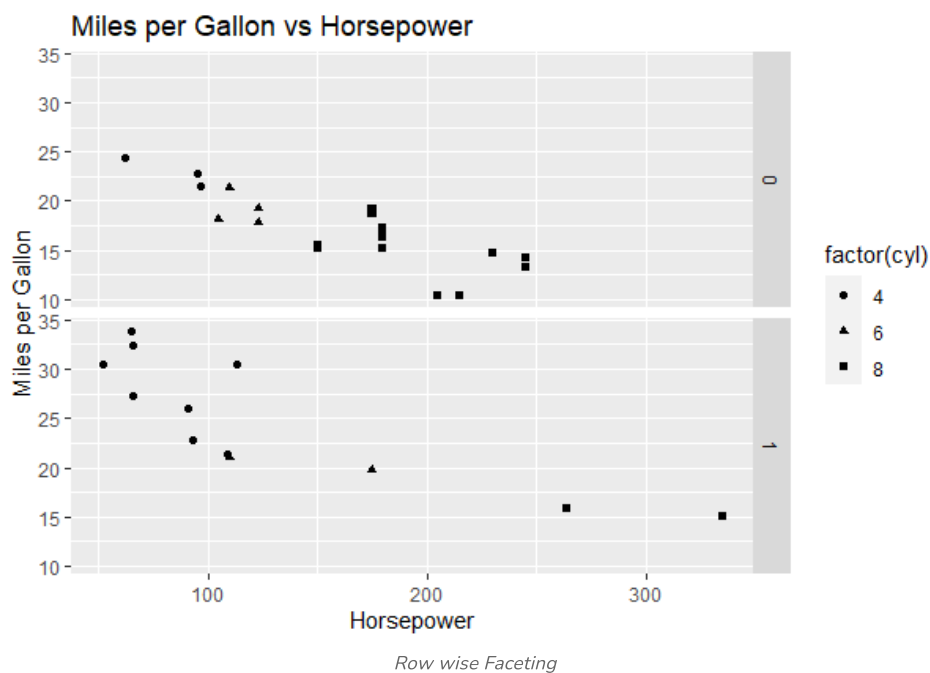
The facet layer is used to split the data up into subsets of the entire dataset and it allows the subsets to be visualized on the same plot. Here we separate rows according to transmission type and Separate columns according to cylinders.

Apply Faceting (Row-wise): Here we split the scatter plot by transmission type.

```
p <- ggplot(data = mtcars, aes(x = hp, y = mpg, shape = factor(cyl))) +
  geom_point()

p + facet_grid(am ~ .) +
  labs(title = "Miles per Gallon vs Horsepower",
        x = "Horsepower",
        y = "Miles per Gallon")
```

Output:



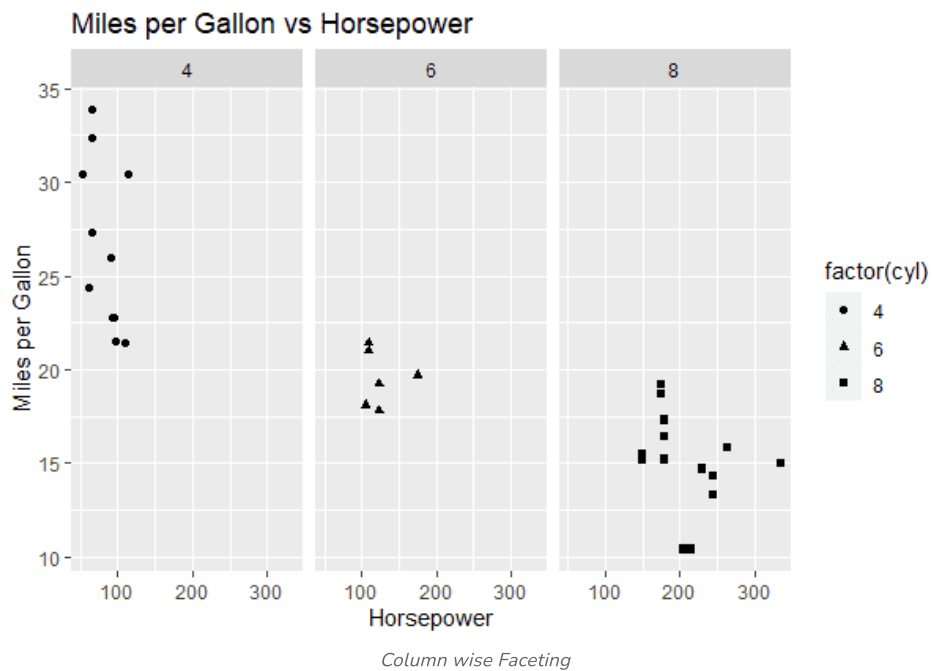
Apply Faceting (Column-wise): Now we split the scatter plot by cylinder count.

```
p <- ggplot(data = mtcars, aes(x = hp, y = mpg, shape = factor(cyl))) +
```

```
geom_point()

p + facet_grid(. ~ cyl) +
  labs(title = "Miles per Gallon vs Horsepower",
        x = "Horsepower",
        y = "Miles per Gallon")
```

Output:

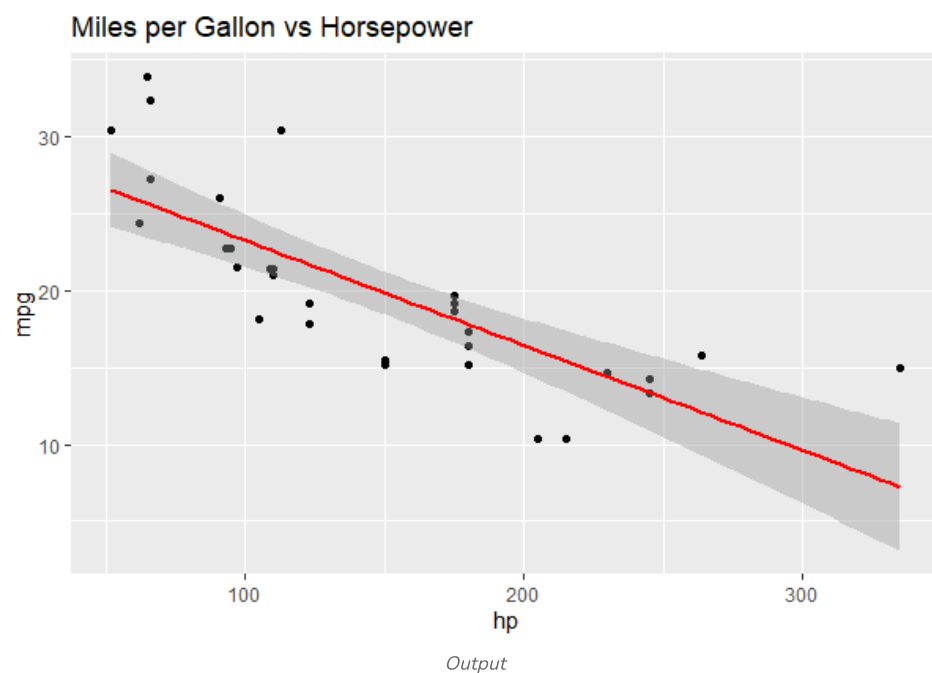


Step 6: Statistics layer

This layer transforms our data using binning, smoothing, descriptive statistics and intermediate summaries. It also adds a linear regression line to show the trend.

```
ggplot(data = mtcars, aes(x = hp, y = mpg)) +
  geom_point() +
  stat_smooth(method = lm, col = "red") +
  labs(title = "Miles per Gallon vs Horsepower")
```

Output:

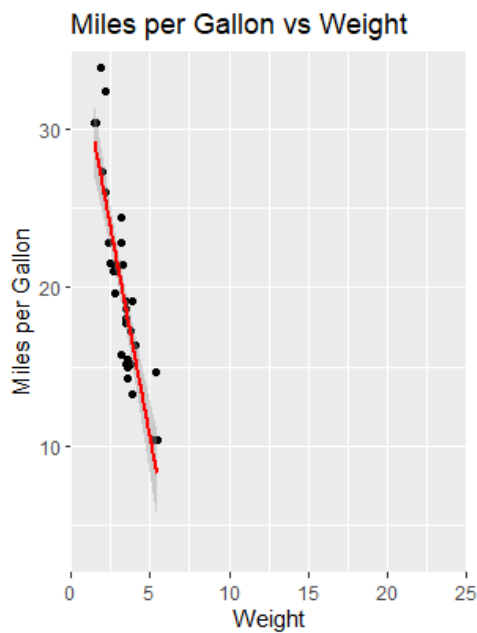


Step 7: Coordinates layer

In these layers, data coordinates are mapped together to the mentioned plane of the graphic and we adjust the axis and changes the spacing of displayed data with Control plot dimensions.

```
ggplot(data = mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  stat_smooth(method = lm, col = "red") +
  scale_y_continuous("Miles per Gallon", limits = c(2, 35), expand = c(0, 0)) +
  scale_x_continuous("Weight", limits = c(0, 25), expand = c(0, 0)) +
  coord_equal() +
  labs(title = "Miles per Gallon vs Weight",
       x = "Weight",
       y = "Miles per Gallon")
```

Output:



Data visualization with R and ggplot2

Zoom Using coord_cartesian():

```
ggplot(data = mtcars, aes(x = wt, y = hp, col = am)) +
  geom_point() +
  geom_smooth() +
  coord_cartesian(xlim = c(3, 6))
```

Output:

